

Time: 3 Hrs.

CTA

Max. Marks: 180

PAPER 2

JEE – ADVANCE – 2020 – P2 – Model

IMPORTANT INSTRUCTIONS

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total Marks
Sec – I(Q.N : 1 – 6)	Questions with Single Digit Integer Type	+3	-1	6	18
Sec – II(Q.N : 7 – 12)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	6	24
Sec – II(Q.N : 13 – 18)	Questions with Numerical Answer Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30 Decimal Numbers)	+3	0	6	18
Total				18	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total Marks
Sec – I(Q.N : 19 – 24)	Questions with Single Digit Integer Type	+3	-1	6	18
Sec – II(Q.N : 25 – 30)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	6	24
Sec – II(Q.N : 31 – 36)	Questions with Numerical Answer Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30 Decimal Numbers)	+3	0	6	18
Total				18	60

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total Marks
Sec – I(Q.N : 37 – 42)	Questions with Single Digit Integer Type	+3	-1	6	18
Sec – II(Q.N : 43– 48)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	6	24
Sec – II(Q.N : 49– 54)	Questions with Numerical Answer Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30 Decimal Numbers)	+3	0	6	18
Total				18	60

Exam Syllabus

PHYSICS	COMPLETE 11TH CLASS SYLLABUS
CHEMISTRY	COMPLETE 11TH CLASS SYLLABUS
MATHEMATICS	COMPLETE 11TH CLASS SYLLABUS

SECTION – I
(SINGLE DIGIT INTEGER TYPE)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from **0 TO 9, BOTH INCLUSIVE**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct integer is entered;
Zero Marks : 0 If the question is unanswered;
Negative Marks : –1 In all other cases.

1. A point is moving along a circular path with a constant tangential acceleration starting from rest at $t = 0$. Ratio of radial to tangential acceleration is 6π at time t . Find the number of rotations completed by the particle upto time t will be $3/n$ rotation. Find value of n ?
2. In a process, the molar heat capacity of a diatomic gas is $\frac{10R}{3}$. When heat Q is supplied to the gas, find the value of n if work done by Q/n ?
3. A Carnot engine working between 300 K and 600 K has a work output of 800 J per cycle. The amount of heat energy supplied to engine from source to each cycle is $800n$ J. find value of n ?
4. A string is stretched between fixed points separated by 75.0 cm. It is observed to have two consecutive resonant frequencies of 315 Hz and 420 Hz. There are no other resonant frequencies between these two. Then the lowest resonant frequency of string is $15n$ Hz. find value of n ?
5. A particle attached to a string of length L is projected vertically upwards with velocity $V = \sqrt{2gl}$. The other end of the string is attached to a fixed support. The angle $\sin \theta = (2/n)$ made by the string with horizontal when horizontal component of velocity is maximum. Then find value of n



6. The temperature of 1 mole of an ideal gas is decreased from 324 K to 300 K adiabatically in which amount of work done on the gas is $60R$. Heat supplied to the gas in increasing the temperature of the gas from 300 K to 324 K under isochoric process is $15nR$ find value of n ?

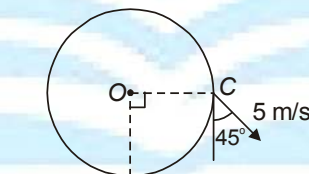
Rough Work

SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -2 In all other cases.

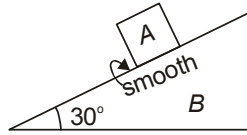
7. At an angle θ with horizontal, a particle is projected from surface of the earth with velocity v_0 to attain the maximum height h (R is radius of earth)
- (A) The speed of the particle at maximum height is $\frac{RV_0 \cos \theta}{(R+h)}$
- (B) For a particle projected from surface of the earth, angular momentum is conserved about centre of earth
- (C) The speed of the particle at maximum height is $\frac{RV_0 \tan \theta}{\sqrt{2}(R+h)}$
- (D) For a particle projected from surface of the earth, angular momentum is not conserved about centre of earth
8. At some instant a disk of mass 1.0 kg and radius 1.0 m is performing rolling on a rough horizontal surface. The velocity of point C is 5 m/s in the direction as shown. Find the correct options.



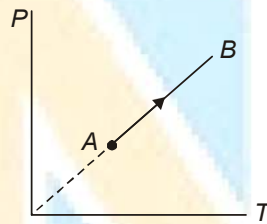
- (A) Instantaneous centre is at a distance 1.0 m below centre O
- (B) Instantaneous centre is at a distance 0.50 m below centre O
- (C) Angular velocity of disc is $\left(\frac{5}{\sqrt{2}}\right)$ rad/s
- (D) Velocity of centre is $\left(\frac{5}{\sqrt{2}}\right)$ m/s

Rough Work

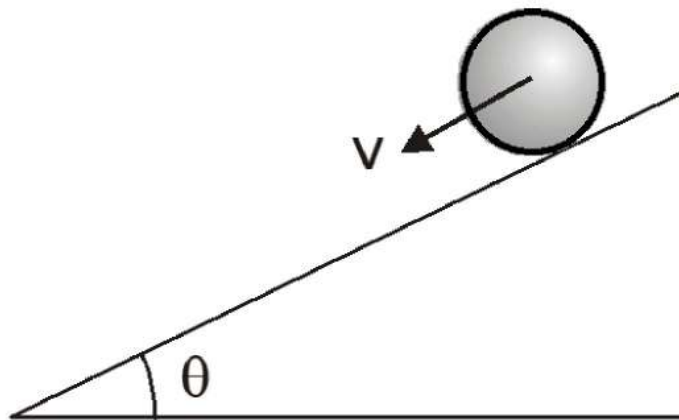
9. Block A is placed on an inclined plane B making an angle of 30° with horizontal. R_2 is the reaction on A when B is stationary. R_1 is the reaction on A when B is moving with constant acceleration towards left



- (A) If $\frac{R_1}{R_2} = \frac{4}{3}$, then block B moves with acceleration $\frac{g}{\sqrt{3}}$
 (B) If $\frac{R_1}{R_2} = \frac{4}{3}$, then block B moves with acceleration $\frac{g}{\sqrt{2}}$
 (C) If $\frac{R_1}{R_2} = \frac{4}{3}$, then block B remains stationary relative to block A
 (D) If $\frac{R_1}{R_2} = \frac{4}{3}$, then block B will not be stationary relative to block A
10. For an ideal gas during process AB , find correct options.



- (A) Density of gas is increasing
 (B) The work done on the gas is zero
 (C) The slope of the line AB is inversely proportional to the volume of gas
 (D) The slope of the line AB is directly proportional to number of moles of the gas
11. A solid sphere is in pure rolling motion on an inclined surface having inclination θ



- (A) Frictional force acting on sphere is $f = \mu mg \cos \theta$
 (B) Friction is dissipative force in this case

Rough Work



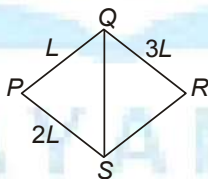
- (C) Friction will increase its angular velocity and decrease its linear velocity
- (D) If θ decreases, friction will decrease
12. A rod of length l is held with one end at $O(0, 0)$ on a smooth horizontal table. When the rod is released it makes angle with vertical
- (A) If $\alpha = 0$ when rod is released, then the centre of mass should move on the line $x = 0$
- (B) If $\alpha = 0$ when rod is released, then the centre of mass should move on line $x - y = 0$
- (C) The angular velocity of rod when it makes angle θ with vertical $\left[\frac{12g(\cos \alpha - \cos \theta)}{\ell(1 + 3\cos^2 \theta)} \right]^{1/2}$
- (D) The angular velocity of rod when it makes an angle θ with vertical is $\left[\frac{12g(\cos \alpha - \cos \theta)}{\ell(1 + 3\sin^2 \theta)} \right]^{1/2}$

SECTION-III

(Numerical Value Answer Type)

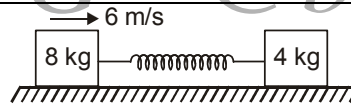
- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered;
Zero Marks : 0 In all other cases.

13. Sand drops vertically at the rate of 5 kg/s on to a conveyor belt moving horizontally with a velocity of 0.6 m/s. Find the extra force required to keep the belt moving (in N)
14. In the arrangement as shown, temperature of end P and R are maintained at 200°C and 0°C respectively. The length of rods are shown. The length of rod RS is nL . If no heat flows through QS , find the value of n . Assume material of the rods and their cross-sectional areas to be same

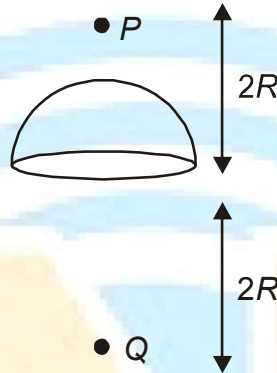


15. Force acting on a particle is $F = 8x$ in SHM. The amplitude of oscillations is 2 (in m) and mass of the particle is 0.5 kg. The total mechanical energy of the particle is 20 J. Find the potential energy of the particle in mean position (in J).
16. Two blocks A and B of masses 8 kg and 4 kg placed on a smooth horizontal surface are connected with a spring. An impulse provides a velocity 6 m/s to block A . The velocity of centre of mass is _____.

Rough Work



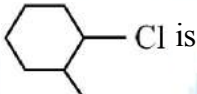
17. A particle of mass m is allowed to oscillate near the minimum of a vertical parabolic path $x^2=4ay$. The angular frequency of small oscillation is $\sqrt{\frac{g}{na}}$ find value of n ?
18. If gravitational field at P due to uniform thin hemispherical shell is I , then the magnitude of field at Q is (Mass of hemisphere is M , radius R) $\frac{GM}{2R^2} - nI$ then find the value of n ?



Rough Work

SECTION – I
(SINGLE DIGIT INTEGER TYPE)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from **0 TO 9, BOTH INCLUSIVE**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct integer is entered;
Zero Marks : 0 If the question is unanswered;
Negative Marks : -1 In all other cases.

19. How many $sp^2 - sp^2$ C–C σ bonds are present in acetophenone?
20. The number of orbitals among the following set of orbitals that have zero probability of finding electron along z axis is(are):
 (a) $3p_z$ (b) $2p_x$ (c) $3d_{xy}$ (d) $4d_{yz}$
 (e) $4s$ (f) $5d_{x^2-y^2}$ (g) $3d_{xz}$ (h) $4p_y$
21. No. of lone pairs in N_2O_3 is x and No. of lone pairs in N_2O_5 is y then $(y - x)$ is
22. If for the reaction at $27^\circ C$, $2A(g) + B(g) \rightleftharpoons 2C(g)$, $\Delta U = 25.6$ kcal and entropy of A, B, C are 200, 100, 500 cal/K mol respectively, then the magnitude of ΔG (in kcal) of reaction is x. The value of $\frac{x}{25}$ is.
23. If 1 mol of HA (weak monobasic acid) is neutralised by 1 mol of NaOH and total heat released is used to raise temperature of 7 mol of ice from $-10^\circ C$ to $+10^\circ C$. If heat of neutralisation of 1 mole of strong acid with of 1 mole of strong base is 57.057 kJ, heat of ionisation of HA (in kJ/mol) is
- $\Delta_{\text{fusion}} H$ of ice = 6.03 kJ mol^{-1}
 $C_p[H_2O(l)] = 75.3 \text{ J mol}^{-1} \text{ K}^{-1}$
 $C_p[H_2O(s)] = 36.8 \text{ J mol}^{-1} \text{ K}^{-1}$
24. Number of chiral carbon atoms in  Cl is

Rough Work

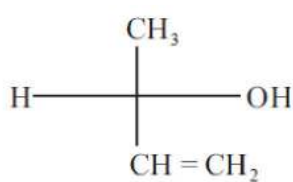
SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

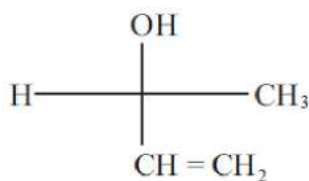
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks.

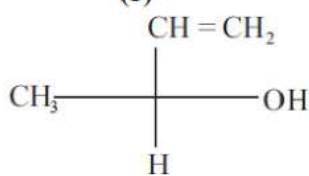
25. Which of the following combinations amongst the four Fischer projections represents the different absolute configuration?



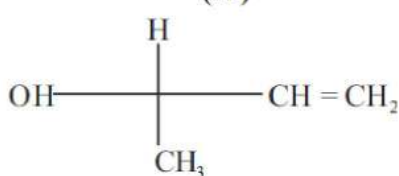
(I)



(II)



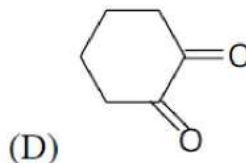
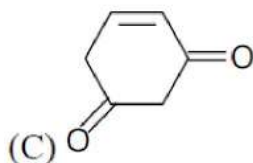
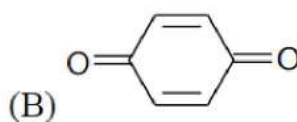
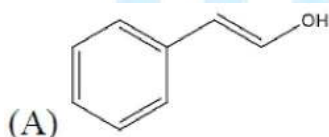
(III)



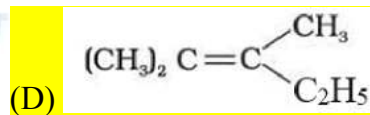
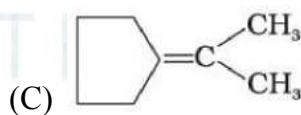
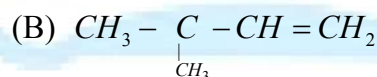
(IV)

- (A) (I) and (II) (B) (I) and (III)
(C) (I) and (IV) (D) (II) and (III)

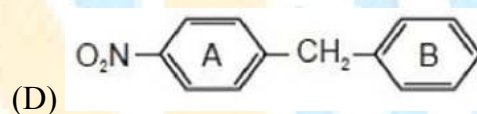
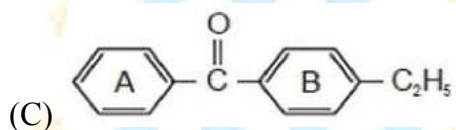
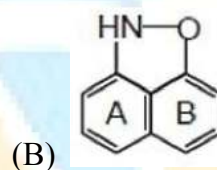
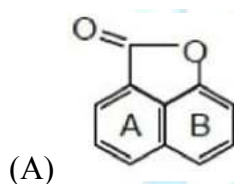
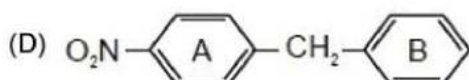
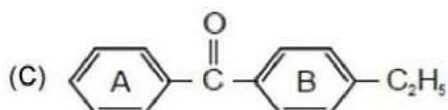
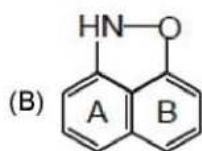
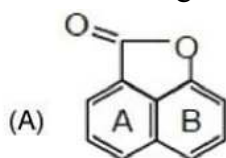
26. Tautomerism is not exhibited by:



27. Which of the following alkenes on ozonolysis give a mixture of ketones only?



Rough Work



29. Which of the following are correct?

(A) $CH_3-O-CH_2^+$ is more stable than $CH_3-CH_2^+$

(B) $(CH_3)_2CH^+$ is more stable than $CH_3-CH_2-CH_2^+$

(C) $CH_2=CH-CH_2^+$ is more stable than $CH_3-CH_2-CH_2^+$

(D) $CH_2=CH^+$ is more stable than $CH_3-CH_2^+$

30. Hyperconjugation involves delocalization of _____.

- (A) electrons of carbon-hydrogen σ bond of an group directly attached to an atom of unsaturated system
 (B) electrons of carbon-hydrogen σ bond of alkyl group directly attached to the positively charged carbon atom.
 (C) π -electrons of carbon-carbon bond
 (D) lone pair of electrons

Rough Work

SECTION-III

(Numerical Value Answer Type)

This section contains 6 questions. The answer to each question is a Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths Place).

eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

Marking scheme: +3 for correct answer, 0 in all other cases.

31. The oxidation state of Cr in CrO_5 is '+x'. What is 'x'?
32. An isobaric process, the ratio of heat supplied to the system and work done by the system for a diatomic gas is a : 2. What is the value of a?
33. In photoelectric effect, the energy of photon striking a metallic surface is $5.6 \times 10^{-19} \text{ J}$. The kinetic energy of the ejected electrons is $12.0 \times 10^{-20} \text{ J}$. The work function is $x \times 10^{-19} \text{ J}$ then x is _____
34. What % of $\text{AB}_{3(g)}$ is dissociated as, $\text{AB}_{3(g)} \rightleftharpoons \text{AB}_{2(g)} + \frac{1}{2} \text{B}_{2(g)}$ when the initial pressure of AB_3 is 800 torr. and total pressure developed at equilibrium is 1000 torr. What fraction of $\text{AB}_{3(g)}$ is dissociated?
35. The first four successive ionization energies for an element are 6.113, 11.87, 50.98 & 67.01 (in eV) respectively. The number of Valence electrons in the given element is _____
36. Among BF_3 , $\text{N}(\text{SiH}_3)_3$, $\text{N}(\text{CH}_3)_3$, $\text{O}(\text{SiH}_3)_2$, CF_3^- , CCl_3^- the number of molecules or molecular ions which have back bonding is _____

Rough Work

MATHEMATICS

Max Marks: 60

SECTION – I (SINGLE DIGIT INTEGER TYPE)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from **0 TO 9, BOTH INCLUSIVE**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct integer is entered;
Zero Marks : 0 If the question is unanswered;
Negative Marks : –1 In all other cases.

37. The coefficient of x^5 in the expansion of $\left(2x^3 - \frac{1}{3x^2}\right)^5$ is $\frac{80}{\lambda}$, then value of λ is
38. If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is $\frac{\lambda}{3}$, then value of λ is.
39. If $x = a + bi$ is a complex number such that $x^2 = 3 + 4i$ and $x^3 = 2 + 11i$, where $i = \sqrt{-1}$ then $(a + b)$ equal to.
40. The complex number z satisfies $z + |z| = 2 + 8i$. The value of $\left(\frac{1}{17}|z|\right)$ is
41. If $x = 2 + 2^{2/3} + 2^{1/3}$ then the value of $x^3 - 6x^2 + 6x$ is.
42. If $x, 2y, 3z$ are in A.P., where the distinct numbers x, y, z are in G.P., and the common ratio of the G.P. is 3λ , then value of λ is

SECTION – II (MULTIPLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks.

43. Indicate the correct alternative(s), for $0 < \phi < \pi/2$, if:

$$x = \sum_{n=0}^{\infty} \cos^{2n} \phi, y = \sum_{n=0}^{\infty} \sin^{2n} \phi, z = \sum_{n=0}^{\infty} \cos^{2n} \phi \sin^{2n} \phi \text{ then:}$$

- (A) $xyz = xz + y$ (B) $xyz = xy + z$ (C) $xyz = x + y + z$ (D) $xyz = yz + x$

Rough Work

44. If $\begin{cases} (2a-5)x^2 - 4x - 15 = 0 \\ (3a-8)x^2 - 5x - 21 = 0 \end{cases}$ have a common root then
 (A) $a = 4$ (B) $a = 6$ (C) $a = 8$ (D) $a = 1$
45. Let x and y be real variables satisfying $x^2 + y^2 + 8x - 10y - 40 = 0$. Let $a = \max \left\{ \sqrt{(x+2)^2 + (y-3)^2} \right\}$ and $b = \min \left\{ \sqrt{(x+2)^2 + (y-3)^2} \right\}$. Then,
 (A) $a + b = 18$ (B) $a + b = \sqrt{2}$
 (C) $a - b = 4\sqrt{2}$ (D) $a \cdot b = 73$
46. If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then
 (A) $\tan^2 x = \frac{2}{3}$ (B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$
 (C) $\tan^2 x = \frac{1}{3}$ (D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$
47. If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ at four points $P(x_1, y_1), Q(x_2, y_2), R(x_3, y_3)$ and $S(x_4, y_4)$, then
 (A) $x_1 + x_2 + x_3 + x_4 = 0$ (B) $y_1 + y_2 + y_3 + y_4 = 0$
 (C) $x_1 x_2 x_3 x_4 = c^4$ (D) $y_1 y_2 y_3 y_4 = c^4$
48. The lines parallel to the normal to the curve $xy = 1$ is/ are
 (A) $3x + 4y + 5 = 0$ (B) $3x - 4y + 5 = 0$
 (C) $4x + 3y + 5 = 0$ (D) $3y - 4x + 5 = 0$

Rough Work

SECTION-III

(Numerical Value Answer Type)

This section contains 6 questions. The answer to each question is a Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths Place).

eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

Marking scheme: +3 for correct answer, 0 in all other cases.

49. If z is any complex number satisfying $|z - 3 - 2i| \leq 2$, then the minimum value of $\frac{1}{2} |2z - 6 + 5i|$ is.
50. Given that $1 + 2|z|^2 = |z^2 + 1|^2 + 2|z + 1|^2$, then the value of $\frac{1}{2} |z(z + 1)|$ is
51. Let ABC be a triangle. Let A be the point (1, 2), $y = x$ be the perpendicular bisector of AB, and $x - 2y + 1 = 0$ be the angle bisector of $\angle C$. If the equation of BC is given by $ax + by - 5 = 0$, then the value of $\frac{1}{5}(a + b)$ is
52. If the variable line $y = kx + 2h$ is tangent to an ellipse $2x^2 + 3y^2 = 6$, then the locus of P(h, k) is a conic C whose eccentricity is e. then the value of $\frac{3}{2}e^2$ is.
53. Let the sum of first three terms of G.P. with real terms be $13/12$ and their product is -1. If the absolute value of the sum of their infinite terms is S, then the value of $\frac{7}{8}S$ is _____.
54. If the second term of the expansion $\left[a^{1/13} + \frac{a}{\sqrt{a^{-1}}} \right]^n$ is $14a^{5/2}$, then the value of $\frac{1}{5} \frac{{}^nC_3}{{}^nC_2}$ is

Rough Work